

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4 spaces indentation followed in Python files
2	Proper File Structure	Files and folders are logically organized	Yes	Organized by modules and features
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive and self-explanatory names used
4	Function / Method Names	Functions are descriptively named	Yes	Followed snake_case naming convention
5	Code Comments	Inline and block comments explain logic clearly	Yes	Important logic has been commented
6	Modular Design	Code is split into reusable functions/modules	Yes	Utilized reusable modules and services
7	No Redundant Code	Duplicate or unused code is removed	Yes	Code reviewed and cleaned
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-except blocks used wherever required

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Authentication Module	Python (FastAPI)	User Login, Token Validation	High
2	Data Preprocessing Module	Python (Pandas)	Data Cleaning, Feature Engineering	High
3	Machine Learning Service	Python (Scikit-learn)	Loan Prediction, Risk Assessment	High
4	Email Notification Service	Python (SMTP)	Registration, Alerts, Updates	Medium
5	Database Connector	Python (SQLAlchemy)	All DB Operations	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-structured code with consistent indentation, proper naming conventions, and modular design.
Readability	5	Code is easy to read and understand. Good use of meaningful names and clear logic.
Reusability	5	Code is divided into reusable modules and functions which can be used in future projects.
Documentation / Comments	4	Sufficient comments are provided. Some complex logic can be further explained.
Overall Score	19 / 20	Excellent code quality. Maintain and continue following good coding practices.